LOTQ-0049 / PYTHON FOR QUANT RESEARCH / INTERMEDIATE / FIELD GUIDE

Vectorized Research Workflows in Python

Build fast array workflows with explicit contracts, causal transforms, and reproducibility.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0049
CATEGORY	Python for Quant Research
DIFFICULTY	Intermediate
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Table of contents

INTERMEDIATE

ORIENTATION

ARRAY THINKING AND CONTRACTS

VECTORIZED TRANSFORMATIONS - A

Table of contents

INTERMEDIATE

VECTORIZED TRANSFORMATIONS - B

WORKFLOW ARCHITECTURE

PERFORMANCE AND RELIABILITY - A

Table of contents

INTERMEDIATE

PERFORMANCE AND RELIABILITY - B

VISUAL LABS AND REVIEW

ORIENTATION

How to use this manual

INTERMEDIATE

1. START FROM SEMANTICS

Write row grain, axis meanings, units, clocks, and eligible population before changing code. A valid vector expression matters only when it calculates the intended research quantity.

2. REPRODUCE A TINY PANEL

Use three dates and four assets, including a gap and a tie. Predict labels, shapes, masks, counts, and numbers before executing the optimized path.

3. SEPARATE LAYERS

Keep data access and labeling outside pure numerical kernels. This isolates identity errors and lets kernels be tested independently of files and notebooks.

4. AUDIT POPULATION CHANGES

Joins, masks, reshapes, and missing policies alter who survives. Record input rows, output rows, unmatched keys, exclusions, and denominators.

5. DRAW TIME

Mark event time, arrival time, feature readiness, decision cutoff, entry, and target interval. Prove windows and fitted transforms implement that information set.

6. MEASURE HONESTLY

Benchmark realistic shapes, dtypes, groups, and missingness. Preserve output checksums and memory peaks so speed cannot conceal a policy change.

7. TEST EQUIVALENT MODES

Run one golden event stream through batch, chunked, and streaming implementations. Compare keys, values, flags, counts, and boundary state.

DECISION STANDARD

Promote only when semantics, causality, numerics, ownership, performance, replay, and lineage are explicit and verified.

REFERENCE SHEET

Notation and array contracts

INTERMEDIATE

FORM	OBJECT	CONTRACT
X[T,N]	**return panel**	T ordered decision times by N ordered assets
F[M,K]	**feature matrix**	M frozen samples by K registered features
y[M]	**target vector**	One outcome aligned to each sample identity
w[N]	**weight vector**	Weights in the same asset order as X
M[T,N]	**validity mask**	Eligibility state at each panel position
c[T]	**valid count**	Effective denominator for each time slice
S	**reducer state**	State carried across chunks or ordered events
t_event	**event time**	When the market event occurred
t_ready	**ready time**	When the derived value became usable
t_cutoff	**decision cutoff**	Latest admissible information instant

READING RULE

Every array travels with axis identities, units, missing policy, clock meaning, and source population. Shape checks protect computation; sample IDs and cutoffs protect research meaning.

PROCESS FLOW

Vectorized research workflow

INTERMEDIATE

1. **Preserve evidence**
   Raw values, IDs, event and arrival clocks, revisions
2. **Canonicalize**
   Long-form grain, keys, types, units, calendars, quality
3. **Build information set**
   Cutoff, point-in-time universe, joins, exclusions
4. **Freeze labeled view**
   Exact sample order, masks, schemas, coverage
5. **Run pure kernels**
   Broadcast, reduce, roll, rank, reshape, matrix math
6. **Validate equivalence**
   Hand fixtures, properties, perturbation, parity
7. **Publish artifact**
   Labels restored, manifest, checksums, lineage
8. **Replay and monitor**
   Freshness, coverage, drift, performance, rollback

BOUNDARY RULE

Identity and time are resolved before positional computation. Returning to the labeled world restores keys, units, flags, counts, and replay evidence.

Vectorization as a data model

INTERMEDIATE

ONE-SENTENCE MEANING

Vectorization expresses one operation over a typed collection instead of dispatching Python code for every element. Its value is a clearer data model plus compiled execution, not short syntax alone.

MEMORY JOGGER

Think in whole arrays with named axes, units, and clocks. Before removing a loop, state which observations may interact and which must remain independent.

WHY IT MATTERS IN QUANT RESEARCH

Returns, scores, exposures, and simulations become faster and easier to test when the population is explicit. A vectorized error also scales across the full dataset, so semantics must precede speed.

FORMULA INTUITION

For elementwise $y = f(x)$, output positions are independent once x is fixed. Reductions, scans, and matrix products add distinct interaction rules that should be documented.

SIMPLE MARKET EXAMPLE

Array slices compute ten million log returns without interpreter dispatch. Positive-price checks and interval timestamps still determine whether those values are valid research evidence.

PYTHON / SYSTEM IMPLEMENTATION

Separate I/O, label alignment, and numerical kernels. Give each kernel a shape, dtype, missing-value, clock, and mutability contract, then test it on a hand-sized fixture.

CONFUSIONS TO AVOID

Compact syntax can hide a wrong axis, future-aware shift, or accidental broadcast. Vectorized does not automatically mean causal, stable, or economically meaningful.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Use vector kernels only after canonical data has been aligned to a decision-time view. Restore labels, units, feature identity, and lineage before publishing results.

RELATED CONCEPTS AND QUICK RECALL

Related: ufuncs, broadcasting, reductions, array contracts. Recall task: Separate the numerical and semantic requirements of a return calculation.

Shape and axis contracts

INTERMEDIATE

ONE-SENTENCE MEANING

A shape contract assigns a domain meaning to every array dimension and states how an operation changes those dimensions. Equal lengths do not prove equal identities.

MEMORY JOGGER

Write T x N x F as time by assets by features, not merely three integers. Every reshape, transpose, reduction, and broadcast should preserve or deliberately transform that sentence.

WHY IT MATTERS IN QUANT RESEARCH

High-impact vectorization bugs are often plausible axis errors: weights applied by date or ranks computed through time. Assertions move these failures to the kernel boundary.

FORMULA INTUITION

For X[T,N], mean(X, axis=0) returns N time averages; mean(X, axis=1) returns T cross-sectional averages. Both outputs may look reasonable while answering different questions.

SIMPLE MARKET EXAMPLE

A 252 x 500 panel standardized on axis 0 creates per-asset histories. Standardizing on axis 1 creates daily relative scores instead.

PYTHON / SYSTEM IMPLEMENTATION

Document named shapes, assert dimensions, and require labeled adapters to freeze order. Test singleton, empty, and transposed inputs explicitly.

CONFUSIONS TO AVOID

squeeze can delete a business axis when its current length is one. reshape can preserve bytes while changing which symbol-date pair occupies each position.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Store logical axes beside serialized arrays. Feature, model, and portfolio stages must validate the same order before positional computation.

RELATED CONCEPTS AND QUICK RECALL

Related: axis, reshape, transpose, dimensional contract. Recall task: Contrast time-series and cross-sectional means for one return panel.

Labels before positions

INTERMEDIATE

ONE-SENTENCE MEANING

Pandas labels protect identity during joins and alignment, while NumPy positions provide compact numerical execution. Cross to positions only after exact identity equality is proved.

MEMORY JOGGER

Labels resolve who and when; arrays resolve how much. Freeze the index, assert equality, compute positionally, and restore identity afterward.

WHY IT MATTERS IN QUANT RESEARCH

Weights, returns, features, and targets often share a length but not an order or universe. Label-first alignment prevents one asset's value from being applied to another.

FORMULA INTUITION

Series multiplication aligns symbols, while two to_numpy arrays use position only. The positional form is valid only after both indexes are unique and exactly equal.

SIMPLE MARKET EXAMPLE

A rebalance has 500 weights but the return panel has 497 securities after delistings. Explicit reindexing exposes the three missing identities before portfolio PnL is calculated.

PYTHON / SYSTEM IMPLEMENTATION

Resolve duplicates, select a population policy, sort by a canonical key, and assert index equality. Save the frozen index with the kernel output and checksum it.

CONFUSIONS TO AVOID

Converting early to arrays removes evidence rather than overhead alone. Sorting two objects independently can still yield different populations with matching lengths.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Use labels at ingestion and artifact boundaries, arrays inside verified kernels, and labeled outputs afterward. The adapter owns conversion evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: Index alignment, canonical universe, positional kernel, reindex. Recall task: State every precondition for safely multiplying two unlabeled vectors.

Dtype and numerical policy

INTERMEDIATE

ONE-SENTENCE MEANING

Dtype controls range, precision, missing representation, and arithmetic behavior for every vectorized operation. Choose types from economic requirements rather than ingestion defaults.

MEMORY JOGGER

The fastest incorrect dtype is still a data defect. Decide precision, overflow behavior, and null representation before optimizing storage.

WHY IT MATTERS IN QUANT RESEARCH

Large sums, compounded returns, covariance matrices, and timestamps are sensitive to downcasting. Object columns also disable optimized kernels and hide schema drift.

FORMULA INTUITION

float32 carries roughly seven decimal digits and float64 roughly sixteen; fixed-width integers can overflow. Critical reductions need an explicit accumulator dtype.

SIMPLE MARKET EXAMPLE

Millions of float32 PnL increments can drift from a float64 reference. A narrow integer quantity can wrap during notional multiplication and remain superficially valid.

PYTHON / SYSTEM IMPLEMENTATION

Validate ranges before casts, use stable precision for research math, and record logical plus physical types. Compare optimized output with a high-precision reference on extremes.

CONFUSIONS TO AVOID

astype may truncate without preserving intent, while object dtype can mix numbers and text. NaN does not explain whether a value is unavailable, invalid, or inapplicable.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Canonicalization assigns governed dtypes; kernels promote deliberately; publication restores schema types after checks. Monitoring tracks null and overflow indicators.

RELATED CONCEPTS AND QUICK RECALL

Related: precision, overflow, nullable types, accumulator. Recall task: Choose dtypes for prices, quantities, flags, and nanosecond timestamps.

Broadcasting with economic meaning

INTERMEDIATE

ONE-SENTENCE MEANING

Broadcasting combines compatible shapes by logically extending singleton dimensions without repeating stored values. It is safe only when each extension matches the intended economic axis.

MEMORY JOGGER

Align shapes from the right, then name the axis each operand addresses. Computational compatibility is not proof of semantic alignment.

WHY IT MATTERS IN QUANT RESEARCH

Demeaning returns, applying costs, scaling features, and shocking scenarios all benefit from broadcasting. Explicit singleton dimensions make the direction reviewable.

FORMULA INTUITION

X[T,N] - mu[N] centers each asset through time, while X[T,N] - daily[T,1] centers each day's cross-section. Both return T x N with different meaning.

SIMPLE MARKET EXAMPLE

A 500-element volatility vector can scale a 252 x 500 panel. If ordered by sector rather than symbol, it broadcasts successfully and is economically wrong.

PYTHON / SYSTEM IMPLEMENTATION

Use named shapes such as asset_scale[None,:] and day_mean[:,None]. Assert output shape and compare selected cells against scalar calculations.

CONFUSIONS TO AVOID

A one-dimensional operand has no intrinsic row or column meaning. Automatic broadcasting can turn a missing axis declaration into a silent model change.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Align labels before raw vectors and reject ambiguous public API arguments. Store axis maps beside every dense kernel input.

RELATED CONCEPTS AND QUICK RECALL

Related: singleton dimension, axis identity, elementwise kernel, alignment. Recall task: Show the shapes for asset-wise and date-wise centering.

Views, copies, and ownership

INTERMEDIATE

ONE-SENTENCE MEANING

A view shares an underlying buffer while a copy owns independent bytes. Workflows need an ownership contract because mutation through a view can alter upstream evidence.

MEMORY JOGGER

Ask who owns the bytes, who may mutate them, and when a copy is justified. Treat inputs as immutable unless a narrow work-buffer contract says otherwise.

WHY IT MATTERS IN QUANT RESEARCH

Rolling slices and scenario blocks can be huge, so blind copying wastes memory. Hidden sharing is worse because experiment order may change results.

FORMULA INTUITION

Basic slicing often returns a view; advanced indexing usually creates a copy. A transpose may be noncontiguous and trigger a later implicit copy.

SIMPLE MARKET EXAMPLE

window = returns[-60:] may share storage with returns. In-place winsorization of window can silently rewrite source history.

PYTHON / SYSTEM IMPLEMENTATION

Use pure functions, mark source arrays read-only where practical, and copy at declared ownership boundaries. Measure peak memory and test input immutability.

CONFUSIONS TO AVOID

A shallow container copy does not copy arrays inside it. Copy-on-Write behavior varies by version and does not legitimize chained assignment.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Artifact inputs remain immutable, scratch buffers remain task-local, and published outputs are sealed with checksums. Environment metadata records behavior-affecting options.

RELATED CONCEPTS AND QUICK RECALL

Related: view, copy, contiguity, Copy-on-Write. Recall task: Prove that a feature function leaves its input array unchanged.

Universal functions and fused expressions

INTERMEDIATE

ONE-SENTENCE MEANING

Universal functions apply compiled elementwise operations with defined broadcasting and dtype rules. Compound expressions can still allocate several full-sized temporary arrays.

MEMORY JOGGER

Vectorize the arithmetic, then inspect data movement. A one-line expression can read and write memory many times.

WHY IT MATTERS IN QUANT RESEARCH

Logs, clipping, comparisons, and finite checks dominate many feature pipelines. Shared kernels improve throughput and centralize numerical policy.

FORMULA INTUITION

For y = log1p(x)/scale, positions are independent and the valid domain is x > -1. out and where can control buffers and masked evaluation.

SIMPLE MARKET EXAMPLE

A signed-log transform over 50 million observations beats a Python loop. It still needs rules for infinity, invalid inputs, and a zero scale.

PYTHON / SYSTEM IMPLEMENTATION

Compose short kernels, profile temporaries, and use out only with unambiguous ownership. Compare with a scalar reference and test domain boundaries.

CONFUSIONS TO AVOID

Suppressing warnings can turn bad inputs into quiet NaNs without a reason. In-place output may destroy an operand still needed later.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Tested ufunc kernels belong in a shared numerical layer. Feature definitions wrap them with labels, fitted parameters, timing, masks, and lineage.

RELATED CONCEPTS AND QUICK RECALL

Related: ufunc, temporary array, out parameter, domain check. Recall task: Identify semantic and memory checks for a signed-log transform.

Boolean masks and rule composition

INTERMEDIATE

ONE-SENTENCE MEANING

A boolean mask turns data-quality, universe, or trading rules into elementwise eligibility. Named component masks preserve diagnostics that one opaque filter discards.

MEMORY JOGGER

Build predicates separately, count them separately, then combine them. Parentheses are part of correctness for array logical operators.

WHY IT MATTERS IN QUANT RESEARCH

Masks govern tradability, sessions, outliers, model validity, and risk breaches. Reason-level counts reveal selection bias and feed failures.

FORMULA INTUITION

eligible = finite & liquid & active combines aligned boolean arrays. x[eligible] shortens the data, while where can preserve the original grid.

SIMPLE MARKET EXAMPLE

A tradability gate requires finite price, positive volume, tight spread, active listing, and fresh quote. The final count alone cannot explain exclusions.

PYTHON / SYSTEM IMPLEMENTATION

Name predicates, assert boolean dtype and exact shape, and retain reason codes before intersection. Keep labels until the array boundary.

CONFUSIONS TO AVOID

Python and/or are not elementwise, and filtering early removes evidence needed for coverage analysis. Missing booleans need an explicit policy.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A versioned eligibility bundle should serve features, models, and portfolios. Monitoring reports each rule's pass rate and overlap.

RELATED CONCEPTS AND QUICK RECALL

Related: predicate, eligibility, reason code, where. Recall task: Design a five-rule tradability mask and its stored diagnostics.

Reductions and denominators

INTERMEDIATE

ONE-SENTENCE MEANING

A reduction collapses an axis into a sum, mean, extreme, quantile, or other summary. Its meaning is determined by axis, eligibility, weights, missing policy, and accumulator precision.

MEMORY JOGGER

Every scalar needs a population sentence. Store numerator, denominator, effective count, and method beside the estimate.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio PnL, breadth, exposures, and evaluation metrics depend on reductions. Silent denominator changes create stable-looking numbers from unstable coverage.

FORMULA INTUITION

For m = sum(w*x)/sum(w), both sums must use the same eligible positions and sign convention. Variance additionally needs a centering rule and ddof.

SIMPLE MARKET EXAMPLE

A sector return based on three traded names is not comparable with one based on fifty without coverage. A zero denominator must become a governed invalid state.

PYTHON / SYSTEM IMPLEMENTATION

Pass axis and dtype explicitly, calculate valid counts, and define all-empty behavior. Reconcile missing, negative-weight, and zero-weight fixtures by hand.

CONFUSIONS TO AVOID

NaN-aware reductions may return plausible values on empty slices. Library defaults for ddof can also create unexplained research-versus-production differences.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Aggregation artifacts carry effective populations and method versions. Monitoring alerts on denominator changes as well as value changes.

RELATED CONCEPTS AND QUICK RECALL

Related: axis reduction, ddof, weighted mean, effective count. Recall task: Define a safe contract for an all-missing or zero-weight slice.

GroupBy without hidden loops

INTERMEDIATE

ONE-SENTENCE MEANING

GroupBy partitions rows by keys and then aggregates, transforms, or applies logic within each group. Efficient workflows use built-ins and make output grain explicit.

MEMORY JOGGER

Ask whether output is one row per group or one value per input row. That answer usually decides between agg and transform.

WHY IT MATTERS IN QUANT RESEARCH

Per-symbol returns, per-date ranks, sector exposures, and session summaries are core operations. Correct keys prevent information from crossing assets or periods.

FORMULA INTUITION

agg reduces group rows; transform broadcasts a group result back; apply permits arbitrary shape and often hides Python callbacks. Order-sensitive work requires a declared sort.

SIMPLE MARKET EXAMPLE

A daily cross-sectional z-score uses groupby(date).transform and retains every symbol. A sector summary uses agg and intentionally changes grain.

PYTHON / SYSTEM IMPLEMENTATION

Sort by required keys, use named aggregations, and assert output grain plus counts. Benchmark callbacks using representative group cardinality.

CONFUSIONS TO AVOID

Group order is not chronological order. apply may look vectorized while executing slow Python code for thousands of small groups.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Feature contracts state grouping keys, sort keys, minimum size, and output grain. Batch and stream reducers reproduce the same semantics.

RELATED CONCEPTS AND QUICK RECALL

Related: aggregation, transform, apply, group grain. Recall task: Choose agg or transform for three market-data tasks.

VECTORIZED TRANSFORMATIONS

Rolling windows and membership

INTERMEDIATE

ONE-SENTENCE MEANING

A rolling calculation maps each output timestamp to an ordered set of prior eligible observations. Window length, endpoints, minimum count, and missing-row policy define the feature.

MEMORY JOGGER

Do not start with rolling(20); start with the twenty expected member IDs at one cutoff. Numerical output is secondary to correct membership.

WHY IT MATTERS IN QUANT RESEARCH

Momentum, realized volatility, volume baselines, and risk estimates depend on rolling windows. Endpoint errors can turn a valid feature into same-bar leakage.

FORMULA INTUITION

A trailing L-observation mean uses the latest L eligible values only under a row-count rule. Calendar windows and observation windows are not equivalent.

SIMPLE MARKET EXAMPLE

After a holiday, a 20-session mean uses twenty trading sessions. With missing symbol rows, rolling(20) may reach farther back in calendar time.

PYTHON / SYSTEM IMPLEMENTATION

Define calendar, closed side, label time, min_periods, and availability lag. Test holidays, gaps, duplicates, and exact-boundary decisions.

CONFUSIONS TO AVOID

Centered windows are future-aware for trading features. Filling gaps before rolling can alter both membership and availability.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry stores window semantics and state requirements. Streaming reducers retain admissible trailing state and replay against batch results.

RELATED CONCEPTS AND QUICK RECALL

Related: rolling, expanding, min_periods, endpoint. Recall task: List the membership contract for a trailing 20-session volatility feature.

Sliding-window views

INTERMEDIATE

ONE-SENTENCE MEANING

A sliding-window view exposes overlapping subarrays through strides without copying every window. It can create a huge logical shape despite a cheap initial view.

MEMORY JOGGER

The view is cheap to create, not necessarily cheap to consume. Logical element count can be window length times the source size.

WHY IT MATTERS IN QUANT RESEARCH

Custom rolling statistics, sequence models, and pattern matching may need explicit window matrices. Views make geometry visible while avoiding duplicate storage.

FORMULA INTUITION

For x[T] and window L, the view has shape (T-L+1,L). Adjacent rows share L-1 source elements and commonly share memory.

SIMPLE MARKET EXAMPLE

One million values with a 252-point view expose about 252 million logical cells. A downstream reduction may scan every cell.

PYTHON / SYSTEM IMPLEMENTATION

Check logical shape, keep the source immutable, and prefer specialized algorithms for simple sums or variance. Verify first and last windows explicitly.

CONFUSIONS TO AVOID

Writing through overlapping views can cause order-dependent corruption. Copying the whole window matrix defeats the memory benefit.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Use window views only inside controlled kernels that truly need members. Publish compact results plus the exact window specification.

RELATED CONCEPTS AND QUICK RECALL

Related: strides, overlapping view, rolling matrix, logical size. Recall task: Estimate a window view's logical size and decide when it is unsuitable.

Cross-sectional ranking

INTERMEDIATE

ONE-SENTENCE MEANING

Cross-sectional ranking maps eligible asset values to an order or percentile at one decision time. Tie rules, missing placement, direction, and universe determine membership.

MEMORY JOGGER

Rank within the decision universe, not within the file. Preserve asset identity through every sort and tie break.

WHY IT MATTERS IN QUANT RESEARCH

Factor portfolios, quantile tests, and neutralization depend on reproducible ranks. Nondeterminism at a cutoff creates turnover and unstable validation.

FORMULA INTUITION

Percentile definitions differ for ties and denominators. Stable top-k selection may use partial selection followed by a secondary-key sort.

SIMPLE MARKET EXAMPLE

For scores [0.8,0.8,0.2,missing,-0.1], two assets share the top value. A symbol key or average-rank policy is required before weights are assigned.

PYTHON / SYSTEM IMPLEMENTATION

Apply eligibility first, group by decision time, select direction and tie method, and retain raw scores. Test boundary ties and shuffled input order.

CONFUSIONS TO AVOID

Ranking through time compares an asset with itself, not with peers. Filling missing scores with zero can insert unavailable assets into the portfolio.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ranking service emits deterministic ranks, bins, tie metadata, and membership. Portfolio construction applies risk and capacity afterward.

RELATED CONCEPTS AND QUICK RECALL

Related: rank, quantile, top-k, stable sort. Recall task: Design a deterministic top-decile rule with ties at the cutoff.

Reshaping long and wide data

INTERMEDIATE

ONE-SENTENCE MEANING

Reshaping moves variables between rows, columns, and index levels while preserving observations only when keys are unique. Long tables favor joins; wide matrices favor kernels.

MEMORY JOGGER

Reshape is a change of geometry, not permission to aggregate. Prove the source key before pivoting.

WHY IT MATTERS IN QUANT RESEARCH

Covariance and matrix models need time-by-asset panels, while data systems store timestamp-symbol rows. Reliable conversion connects both without losing identity.

FORMULA INTUITION

pivot requires unique index-column pairs; pivot_table applies an aggregation; melt reverses columns to rows. Nonmissing counts must reconcile across forms.

SIMPLE MARKET EXAMPLE

A long return table pivots to date by symbol for covariance. Averaging duplicate revisions during pivot invents a policy instead of resolving the revision.

PYTHON / SYSTEM IMPLEMENTATION

Assert key uniqueness, freeze column order, and round-trip a fixture. Carry the universe index beside the matrix.

CONFUSIONS TO AVOID

A successful pivot does not prove causal validity. Wide panels can create large sparse arrays and hide field-level metadata.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Canonical storage remains long; jobs materialize versioned matrix views; kernels return labeled outputs. The manifest records axes and source snapshot.

RELATED CONCEPTS AND QUICK RECALL

Related: pivot, melt, stack, matrix view. Recall task: Specify a round-trip test that detects duplicate loss and order changes.

Feature matrices and target alignment

INTERMEDIATE

ONE-SENTENCE MEANING

A model matrix joins features and targets at a precise sample grain while respecting availability and horizons. Vectorized assembly must preserve row identity through every filter.

MEMORY JOGGER

The row is the sample. Freeze sample IDs before producing X and y, then prove both arrays use exactly that order.

WHY IT MATTERS IN QUANT RESEARCH

One misalignment can yield strong but meaningless predictive results because shuffled targets retain realistic distributions. Explicit identities make mapping reviewable.

FORMULA INTUITION

For X[M,F] and y[M], sample_id[M] binds rows. Every mask must be applied to all three objects or realignment must occur by label.

SIMPLE MARKET EXAMPLE

A next-day target removes the final date per asset while feature gaps remove other rows. Building X and y independently can produce equal lengths with different samples.

PYTHON / SYSTEM IMPLEMENTATION

Create one labeled modeling table, apply one eligibility mask, freeze IDs, then extract arrays. Store IDs and split labels with predictions.

CONFUSIONS TO AVOID

Dropping NaNs independently changes populations. Resetting indexes before target alignment destroys evidence needed to catch the mismatch.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dataset builder publishes X, y, sample IDs, feature schema, target definition, cutoffs, splits, and exclusions as one artifact.

RELATED CONCEPTS AND QUICK RECALL

Related: sample ID, design matrix, target horizon, eligibility. Recall task: State an invariant proving predictions map back to the correct samples.

WORKFLOW ARCHITECTURE

Batch, chunk, and stream equivalence

INTERMEDIATE

ONE-SENTENCE MEANING

Equivalent implementations produce the same governed output from the same ordered evidence within declared tolerances. Chunked and live paths need explicit boundary state.

MEMORY JOGGER

One formula, three execution modes, one evidence contract. Boundary rows are where equivalence usually breaks.

WHY IT MATTERS IN QUANT RESEARCH

Research favors batch arrays, large datasets require chunks, and live systems update incrementally. Divergent semantics create backtests that production cannot reproduce.

FORMULA INTUITION

Associative reductions combine summaries; rolling L-observation features carry L-1 eligible members; recursive machines carry full state. Overlap is emitted once.

SIMPLE MARKET EXAMPLE

A 20-bar mean processed monthly needs nineteen prior eligible bars. A live reducer must use the same minimum-count rule before emission.

PYTHON / SYSTEM IMPLEMENTATION

Define ordered inputs, state serialization, overlap, checkpoints, and tolerances. Replay one golden stream through all modes and compare keys, flags, counts, and values.

CONFUSIONS TO AVOID

Equal final values do not prove equal timing or state. Parallel floating reductions may differ, so tolerances require scale-aware justification.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Schedulers own partitions and checkpoints; reducers own mathematical state; parity tests block promotion when modes diverge.

RELATED CONCEPTS AND QUICK RECALL

Related: chunking, streaming reducer, overlap, replay. Recall task: Specify carry state for rolling mean, EWM, and trailing stop.

Immutable artifacts and lineage

INTERMEDIATE

ONE-SENTENCE MEANING

An immutable artifact binds values to source snapshot, schema, code, configuration, environment, and validation evidence. Reproducibility makes speed operationally useful.

MEMORY JOGGER

Store the recipe and ingredients with the dish. A Parquet file alone is not a research result.

WHY IT MATTERS IN QUANT RESEARCH

Pipelines are rerun, parallelized, and revised, so notebook state cannot explain provenance. Content identity enables caching and protects approved evidence.

FORMULA INTUITION

An artifact ID can hash normalized inputs, code, config, and schema; output checksums verify bytes. Lineage forms a graph from evidence to decision.

SIMPLE MARKET EXAMPLE

A feature matrix records bars snapshot, universe, calendar, transform state, definitions, exclusions, and checksums. A vendor correction creates a new artifact.

PYTHON / SYSTEM IMPLEMENTATION

Publish atomically through a manifest, validate counts and checksums, and separate temporary work from durable evidence. Use immutable versioned paths.

CONFUSIONS TO AVOID

A timestamped filename does not prove which data or code produced content. A mutable latest pointer cannot be authoritative evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The artifact store hands work from data to features, models, backtests, and monitoring. Registries reference immutable IDs.

RELATED CONCEPTS AND QUICK RECALL

Related: manifest, checksum, lineage graph, atomic publish. Recall task: List the minimum fields required to reproduce a model matrix.

WORKFLOW ARCHITECTURE

Pure transformation functions

INTERMEDIATE

ONE-SENTENCE MEANING

A pure transformation returns output determined only by explicit inputs and does not mutate external state. This makes kernels composable, replayable, and easy to compare.

MEMORY JOGGER

Inputs in, outputs out, with no hidden clock, global DataFrame, random state, or file read. Side effects belong at orchestration boundaries.

WHY IT MATTERS IN QUANT RESEARCH

Notebooks often mix access, filtering, formulas, plots, and mutation. Pure layers isolate numerical correctness from environment and execution order.

FORMULA INTUITION

A pure f(x,config) returns the same result for immutable inputs and configuration. Determinism also requires controlled libraries and injected random generators.

SIMPLE MARKET EXAMPLE

A winsorization kernel receives values and training bounds, then returns values plus flags. It does not read today's date or refit implicitly.

PYTHON / SYSTEM IMPLEMENTATION

Pass arrays, masks, and fitted parameters explicitly; return typed results; test input immutability. Keep logging metadata outside the mathematical return.

CONFUSIONS TO AVOID

Pure does not guarantee bitwise deterministic parallel reductions. A function reading a module-level cache still has hidden state.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Impure adapters surround a pure transformation graph. The same graph can run in research, batch, replay, and live services.

RELATED CONCEPTS AND QUICK RECALL

Related: referential transparency, dependency injection, immutable input, kernel. Recall task: Refactor a notebook cell into adapter, kernel, and publication stages.

WORKFLOW ARCHITECTURE

Fitted transforms inside validation

INTERMEDIATE

ONE-SENTENCE MEANING

Any transform learning means, scales, bounds, imputations, categories, or bases is part of the model. Fit it on training data and apply it unchanged to validation and test data.

MEMORY JOGGER

If a step calls fit, its state belongs inside the fold. Vectorized preprocessing over the full table is still leakage.

WHY IT MATTERS IN QUANT RESEARCH

Cross-sectional normalization may be contemporaneous, while time-series scaling learns history. Distinguishing them prevents future distribution information entering past samples.

FORMULA INTUITION

For split s, theta_s = fit(X_train_s); transform train and validation with theta_s. Test observations never update that state.

SIMPLE MARKET EXAMPLE

A robust scaler fit over the full decade knows future medians and tail bounds. Even a small numerical effect invalidates the simulated decision process.

PYTHON / SYSTEM IMPLEMENTATION

Clone pipelines per fold, persist state with cutoffs, and add future-perturbation tests. Keep transform fitting separate from application.

CONFUSIONS TO AVOID

Calling transform does not prove the state was fit correctly. Reusing one mutable scaler across parallel folds can cross-contaminate results.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment runner owns fold isolation and registers every state. Production loads approved state under a separate retraining policy.

RELATED CONCEPTS AND QUICK RECALL

Related: fit/transform, fold isolation, preprocessing leakage, state. Recall task: Classify five preprocessing steps as fitted or stateless.

Measure before optimizing

INTERMEDIATE

ONE-SENTENCE MEANING

Performance work starts with representative wall time, CPU, peak memory, allocations, and I/O. Accept an optimization only when semantic and numerical parity pass.

MEMORY JOGGER

Find the bottleneck at production scale and preserve a correctness reference. Tiny benchmarks often optimize the wrong layer.

WHY IT MATTERS IN QUANT RESEARCH

Research latency limits iteration and production latency affects freshness. Neither justifies changing the output contract.

FORMULA INTUITION

Throughput equals processed observations divided by elapsed time. Peak memory includes inputs, temporaries, outputs, and concurrent workers.

SIMPLE MARKET EXAMPLE

Broadcasting cuts CPU but creates three 8-GB temporaries and triggers swapping. A bounded loop may be faster end to end.

PYTHON / SYSTEM IMPLEMENTATION

Use realistic rows, groups, missingness, and storage. Declare cache warm-up, isolate I/O when needed, repeat runs, and checksum outputs.

CONFUSIONS TO AVOID

Notebook timing includes hidden caches and prior state. Faster code may also use different ties, dtypes, or missing policies.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Performance reports define hardware, environment, and regression budgets for time, memory, and artifact size.

RELATED CONCEPTS AND QUICK RECALL

Related: benchmark, throughput, peak memory, regression budget. Recall task: Design a benchmark separating CPU improvement from cache and I/O effects.

Memory layout and temporary arrays

INTERMEDIATE

ONE-SENTENCE MEANING

Strides map logical indexes to byte offsets, while compound expressions may create full-sized temporaries. Data movement can dominate arithmetic in large panels.

MEMORY JOGGER

Count full-sized reads and writes, not lines of code. A transpose may be a cheap view until a kernel forces a copy.

WHY IT MATTERS IN QUANT RESEARCH

Return panels and scenario cubes are often memory-bandwidth bound. Layout, temporaries, and worker concurrency determine whether vectorization scales.

FORMULA INTUITION

C order stores the last axis contiguously; y=(x-mu)/scale may allocate centered and result arrays. Both layout and scratch multiplier affect peak memory.

SIMPLE MARKET EXAMPLE

Three float64 temporaries over 200 million cells require about 4.8 GB beyond the source. Parallel workers multiply that peak.

PYTHON / SYSTEM IMPLEMENTATION

Inspect strides and flags, profile allocations, standardize layout at one boundary, and chunk when safe. Use in-place buffers only with clear ownership.

CONFUSIONS TO AVOID

Contiguous is not optimal for every access pattern. Premature mutation can corrupt shared evidence, while copying may cost more than strided access.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Kernels declare input order and scratch requirements. Schedulers cap concurrency by measured peak memory rather than source size.

RELATED CONCEPTS AND QUICK RECALL

Related: strides, C order, temporary, memory bandwidth. Recall task: Estimate peak memory and choose a layout for a cross-sectional kernel.

PERFORMANCE AND RELIABILITY

Path dependence and state machines

INTERMEDIATE

ONE-SENTENCE MEANING

Path-dependent output relies on prior state produced by earlier events, so naive elementwise vectorization is invalid. A chronological state machine is often clearer and safer.

MEMORY JOGGER

If event t changes the rules for event t+1, expose state. Do not force an opaque cumulative trick merely to remove a loop.

WHY IT MATTERS IN QUANT RESEARCH

Stops, fills, inventory, margin, and queues depend on event order. Independent-row logic can create trades after exit or use nonexistent inventory.

FORMULA INTUITION

State evolves as $s_t = F(s_{t-1}, event_t)$; only associative scans can be reorganized safely. Transition records should include prior and resulting state.

SIMPLE MARKET EXAMPLE

A trailing stop updates the peak only while a position is open. Computing all hypothetical breaches may select one after the actual exit.

PYTHON / SYSTEM IMPLEMENTATION

Vectorize preprocessing, then run a typed transition loop. Validate hand paths, gaps, ties, and terminal states before compiling it.

CONFUSIONS TO AVOID

A Python loop is not automatically defective. GroupBy apply can hide the same loop with less visible state and more overhead.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The backtest engine owns event ordering and serialized state. Replay artifacts preserve transitions for PnL and incident analysis.

RELATED CONCEPTS AND QUICK RECALL

Related: recurrence, state machine, scan, event order. Recall task: Classify four calculations as stateless, reducible, or path dependent.

Parallelism and determinism

INTERMEDIATE

ONE-SENTENCE MEANING

Parallelism overlaps independent tasks or splits compatible computations across workers. Deterministic research also needs stable partitions, streams, reductions, and output assembly.

MEMORY JOGGER

Parallelize independent units with explicit identities. More workers can worsen memory, I/O, and numerical ordering.

WHY IT MATTERS IN QUANT RESEARCH

Folds, assets, scenarios, and trials may run concurrently. Reproducibility matters because run-to-run noise obscures genuine strategy differences.

FORMULA INTUITION

Independent tasks share no mutable state; floating reductions may vary by order. Child random streams derived from stable keys prevent duplicated draws.

SIMPLE MARKET EXAMPLE

Eight workers fit eight folds with stable fold IDs and streams. Results are assembled by ID rather than completion order.

PYTHON / SYSTEM IMPLEMENTATION

Pin thread counts, cap worker memory, derive streams from stable keys, and sort outputs before hashing. Test nested BLAS oversubscription.

CONFUSIONS TO AVOID

Forked workers may inherit identical random state or copy huge pages. Exact equality may be unrealistic, but undocumented tolerance is unacceptable.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The orchestrator owns scheduling, stream hierarchy, environment, and deterministic aggregation. Artifacts record worker mappings.

RELATED CONCEPTS AND QUICK RECALL

Related: parallel worker, random stream, reduction order, oversubscription. Recall task: Design deterministic parallel cross-validation with independent streams.

Validation, replay, and monitoring

INTERMEDIATE

ONE-SENTENCE MEANING

A reliable workflow validates inputs and outputs, replays approved evidence, and monitors live distributions plus system health. Correctness is an ongoing contract.

MEMORY JOGGER

Validate meaning, replay decisions, and monitor values plus denominators. Every alert needs an owner and response.

WHY IT MATTERS IN QUANT RESEARCH

Data revisions, upgrades, calendar changes, and live gaps can alter vectorized results. Layered controls catch drift before capital is affected.

FORMULA INTUITION

Checks include unique keys, monotonic entity time, finite accepted values, coverage, parity, checksums, and future-perturbation invariance.

SIMPLE MARKET EXAMPLE

A feature mean stays stable while coverage falls from 98% to 35%; value-only monitoring misses the outage. Replay should reproduce keys, flags, and values.

PYTHON / SYSTEM IMPLEMENTATION

Attach reports to artifacts, keep golden streams, pin dependencies, and run shadow comparisons before upgrades. Define stop, quarantine, fallback, and rollback.

CONFUSIONS TO AVOID

Snapshot approval can bless unexplained drift. Monitoring final PnL alone cannot localize whether data, features, or portfolio logic failed.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Promotion gates span schema, temporal, numerical, parity, performance, and lineage evidence. Live checks reuse the same invariants.

RELATED CONCEPTS AND QUICK RECALL

Related: replay, drift, validation gate, rollback. Recall task: Design a promotion gate and live response for a core return feature.

BLOCK DIAGRAM

Shape map for a portfolio kernel

INTERMEDIATE

QUESTION

How do labeled returns, weights, and portfolio output move through a safe vector boundary? Identity checks remain separate from numerical multiplication.

INVARIANT

Weight symbols equal return columns exactly; output dates equal return rows exactly; units reconcile to portfolio return.

ANNOTATED CHART

Benchmarking execution paths

INTERMEDIATE

SYNTHETIC WORKLOAD

Compute a clipped return transform over increasing panel sizes. Values illustrate scaling; real decisions require production shapes, dtypes, and hardware.

INTERPRETATION

Vector execution reduces dispatch. Bounded chunks may be slower at moderate sizes yet remain stable when a monolithic expression creates excessive temporaries.

ACCEPTANCE EVIDENCE

Record timing distribution, CPU, peak memory, allocations, checksums, warm-up policy, hardware, thread count, and library versions. Reject semantic differences.

HEATMAP

Logical size and memory pressure

INTERMEDIATE

	1M cells	5M cells	20M cells	100M cells
source	8 MB	40 MB	160 MB	800 MB
+1 temp	16 MB	80 MB	320 MB	1,600 MB
+3 temp	32 MB	160 MB	640 MB	3,200 MB
window x20	160 MB	800 MB	3,200 MB	16,000 MB

READING THE MAP

A float64 source uses eight bytes per cell. Temporaries multiply physical memory, while a window view may allocate little yet expose a far larger logical scan.

SCHEDULING IMPLICATION

Cap worker concurrency by measured task peak. Leave headroom for labels, the interpreter, filesystem cache, and output serialization.

CONTROL

Every kernel declares input bytes, scratch multiplier, output bytes, and chunkability. Memory regressions block release even when median runtime improves.

WINDOW TIMELINE

Rolling membership and decision timing

INTERMEDIATE

DECISION

The feature is evaluated after Friday close for a Monday-open decision. Only values and corrections available by Friday cutoff may enter state.

THREE-MEMBER WINDOW

Expected members are Wednesday, Thursday, and Friday sessions. Store their IDs in a golden fixture; the final mean cannot prove membership.

MISSING THURSDAY

A row-count rule may reach back to Tuesday, while a three-session rule may retain Thursday as missing and fail coverage. Choose deliberately.

ENDPOINT

A pre-close Friday decision cannot use Friday's completed return. The same date row may be valid for a later post-close decision.

MINIMUM COUNT

Allowing two observations emits count=2 and a quality flag. A risk control can still require all three.

CHUNK BOUNDARY

A partition carries prior eligible members, not merely physical rows. Overlap outputs are reconciled by stable sample IDs.

STREAMING PARITY

The live reducer updates only when an event becomes available. Replay must match batch members, values, counts, and ready times.

TEST SET

Include a holiday, gap, late correction, duplicate timestamp, and exact-boundary decision. Fixtures specify member IDs and reasons.

WORKED TABLE

Worked cross-sectional ranking

INTERMEDIATE

SYMBOL	SCORE	STATE	AVG RANK	RESULT
AAA	0.80	eligible	1.5	top tie
BBB	0.80	eligible	1.5	top tie
CCC	0.20	eligible	3.0	middle
DDD	-	missing	-	excluded
EEE	-0.10	eligible	4.0	bottom

POPULATION

Ranks use four eligible assets; DDD is unavailable and receives no neutral score. Coverage 4/5 travels with the result.

TIE POLICY

Average ranks give AAA and BBB 1.5. A top-one portfolio still needs a deterministic secondary rule such as symbol or expected cost.

DIRECTION

Higher is better, so rank 1 is strongest. Reversing direction is a named feature change, not an incidental ascending default.

ORDER-INVARIANCE

Shuffle input rows repeatedly, then restore symbol order. Scores, ranks, flags, and selected membership must remain identical.

SYSTEM HANDOFF

Publish decision time, universe ID, raw score, eligibility, rank definition, tie policy, and selected membership before risk constraints.

STATE-MACHINE MAP

When a state machine beats vectorization

INTERMEDIATE

STATELESS TRANSFORM

Return clipping maps each valid input independently and vectorizes naturally. It still needs a domain, mask, and timing contract.

REDUCTION

A portfolio dot product combines assets but has no event-to-event state. A verified matrix product is clear and efficient.

ASSOCIATIVE SCAN

Cumulative wealth and running maximum have ordered dependence with known scans. Initialization and missing policy remain explicit.

RECURSIVE STATE

A trailing stop depends on prior position, running peak, fills, and event order. Model those transitions directly.

EVENT ORDERING

Ties require sequence or venue rules. Timestamp sorting alone can change fills when events share an instant.

HYBRID PATH

Vectorize normalization and candidate triggers, run a typed loop over transitions, then vectorize diagnostics.

ACCELERATION

Compile the isolated loop only after a readable reference passes hand paths and replay. Measure cold and warm execution.

AUDIT OUTPUT

Persist prior state, event identity, action, resulting state, and reason. This explains both PnL and rejected orders.

PARTITION TIMELINE

Chunk boundaries and carried state

INTERMEDIATE

PARTITION PLAN

Monthly files are operational inputs; symbol and session define domain order. Preserve each entity's event sequence.

ROLLING CARRY

A 60-observation statistic carries 59 prior eligible values per symbol. Rejected physical rows are not members.

EWM CARRY

Carry prior value, weight normalization, count, and last ready time. Short overlap can initialize a different series.

CROSS-SECTIONAL BARRIER

A daily rank waits for all expected date partitions or a universe watermark. Symbol workers need a date combine stage.

OVERLAP RECONCILIATION

Identify overlap outputs by stable sample keys and emit once. Dropping by row number is unsafe after filtering.

RETRY SAFETY

Write a temporary partition and commit after validation. The same input and state key produce the same artifact ID.

GLOBAL CHECKS

Counts sum to expected totals, boundaries match one-pass reference, and missing partitions block the final manifest.

RESOURCE POLICY

Chunk size balances I/O, vector efficiency, and memory. Worker count follows measured scratch multipliers.

COMPARISON TABLE

Batch, chunk, and stream equivalence

INTERMEDIATE

INPUT IDENTITY

All modes consume the same immutable evidence IDs in domain order. Stream arrival is replayed from recorded ready times.

POPULATION

Keys, exclusions, and counts match at every cutoff. Equal values with different members fail parity.

NUMERICAL VALUES

Use exact comparison for IDs and flags; justify scale-aware tolerances for floating reductions.

BOUNDARY STATE

Rolling members, EWM state, open positions, and random-stream position match around every partition edge.

AVAILABILITY

Feature ready times and emission order match causality. Batch output cannot appear early because history is loaded.

FAILURES

Invalid records, late corrections, and duplicates produce the same quarantine or correction behavior.

PERFORMANCE

Report throughput, memory, lag, and checkpoint cost. Equivalence gates; performance chooses among passing paths.

EVIDENCE

Store input manifest, state snapshots, checksums, differences, tolerance policy, code, and environment.

DIAGNOSTIC TABLE

Vectorization failure modes and fixes

INTERMEDIATE

WRONG AXIS

Plausible output answers the wrong question. Name axes, assert shape, and compare selected cells by hand.

SILENT BROADCAST

Output shape is correct but entities are wrong. Align labels, add singleton dimensions, and reject ambiguous vectors.

EARLY LABEL LOSS

Equal-length arrays create impossible exposures. Freeze a canonical index and assert equality before conversion.

POPULATION DRIFT

A metric stays stable while its denominator changes. Publish coverage, reasons, and effective counts.

FUTURE LEAKAGE

Backtest collapses live. Define availability, fit transforms inside folds, and perturb future evidence.

TEMPORARY EXPLOSION

Small tests are fast but scale swaps or crashes. Profile allocations, chunk safely, and cap concurrency.

HIDDEN PYTHON LOOP

A concise group operation remains slow. Use built-ins or isolate a typed state kernel.

BOUNDARY MISMATCH

Features jump at file edges. Carry mathematical state and compare against one-pass output.

PARALLEL NONDETERMINISM

Runs vary without explanation. Stabilize partitions, streams, assembly order, and tolerances.

VALIDATION PYRAMID

Testing ladder for vector workflows

INTERMEDIATE

HAND EXAMPLE

Use a tiny labeled panel with a gap, tie, and boundary. Verify member IDs and counts before values.

SCHEMA CONTRACT

Assert fields, dtypes, keys, order, units, clocks, and missing states at each public boundary.

REFERENCE PARITY

Compare the optimized kernel with a simple scalar or labeled implementation over randomized fixtures.

PROPERTIES

Test permutation invariance, rank monotonicity, wealth reconciliation, finite policy, and valid broadcasts.

TEMPORAL PERTURBATION

Change every event after a cutoff. No earlier feature, mask, count, or fitted state may change.

EXECUTION PARITY

Compare batch, chunks, and streams on one ordered sequence, including state near boundaries.

ARTIFACT REPLAY

Rebuild from the manifest in a clean environment and verify keys, checksums, and tolerances.

PERFORMANCE REGRESSION

Measure wall time, peak memory, allocations, and artifact size only after semantic tests pass.

SYSTEM DIAGRAM

Production-grade vector pipeline

INTERMEDIATE

1	**Evidence store**	Events, revisions, IDs, event and ready clocks
2	**Canonical tables**	Grain, types, calendars, quality, stable keys
3	**Point-in-time builder**	Cutoff-valid joins, universe, masks, sample IDs
4	**Labeled transform graph**	Groups, windows, fitted state, definitions
5	**Array adapters**	Frozen order, dtypes, ownership, shape assertions
6	**Numerical kernels**	Broadcast, reduce, rank, matrix, simulate
7	**Artifact publisher**	Labels, manifest, checksums, validation
8	**Batch and live consumers**	Models, backtests, risk, execution, monitoring
9	**Replay control**	Parity, drift, incidents, rollback

CONTROL PRINCIPLE

Positional speed stays inside a verified core. Identity, time, quality, fitted state, and lineage remain visible in every durable output.

RELEASE GATE

Promotion checklist for vectorized research

INTERMEDIATE

SEMANTICS

Row grain, axes, units, populations, grouping keys, sort order, and output grain are asserted.

CAUSALITY

Event, ready, cutoff, entry, and outcome times satisfy the information set; fit state is train-only.

NUMERICS

Dtypes, domains, accumulators, missing behavior, denominators, ddof, and tolerances are explicit.

IDENTITY

Labels persist until exact alignment; sample and asset order are frozen and restored afterward.

OWNERSHIP

Inputs are unchanged, views and copies are deliberate, and scratch buffers cannot escape.

EQUIVALENCE

Hand reference, properties, perturbation, execution parity, and clean replay all pass.

PERFORMANCE

Wall time, CPU, memory, allocations, I/O, and concurrency meet budgets without policy changes.

PUBLICATION

Manifest binds data, schema, calendar, code, config, state, environment, counts, and checksums.

OPERATIONS

Monitoring covers freshness, coverage, drift, parity, latency, resources, fallback, and rollback.

FINAL REVIEW

Final active recall and system index

INTERMEDIATE

ARRAY THINKING

Explain axes, broadcasting, reductions, ownership, dtype, missingness, and the safe point for label removal.

TRANSFORMATIONS

Choose masks, groups, windows, reshapes, matrix products, rankings, and partial selection from contracts.

CAUSALITY

Draw one feature's information set and identify fitted state, endpoints, revisions, and target boundary.

ARCHITECTURE

Trace evidence through canonical data, point-in-time view, labeled transforms, kernels, and artifacts.

SCALE

Estimate source, temporary, output, and concurrent memory; choose layout and chunks that preserve meaning.

STATE

Separate elementwise work, reductions, scans, and recursive machines; state why execution differs.

TESTING

Provide a fixture, invariant, perturbation, parity test, replay test, and performance budget.

OPERATIONS

State monitored value, denominator, freshness, drift, latency, owner, fallback, and rollback.

DECISION STANDARD

Reject speedups whose identity, time, population, numerics, state, or lineage cannot be reconstructed.